

My solutions book doesnt cover the steps they took to prove that the  $\det(A) \neq 0$ . I came to the same conclusion but my answer was wrong, and I simply want to see where my algebra went wrong. So please solve this system. based on the vectors as such. The question is

Use determinants to decide if the set of vectors is linearly dependent

item[25. ] Use determinants to decide if the set of vectors is linearly dependent

$$\begin{bmatrix} \begin{bmatrix} 7 \\ -4 \\ -6 \end{bmatrix}, \begin{bmatrix} -8 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ -5 \end{bmatrix} \end{bmatrix}$$

in the textbook solution, they found its equal to -1.

**TLDR:** Form the  $3 \times 3$  matrix with the vectors as columns, compute the determinant carefully. The correct determinant is  $-1$ . If you got 0 or something else, the mistake is

almost always a sign or minor error. Steps below show exactly where that happens.

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### Step 1: Form the matrix

Use the vectors as **columns** (order doesn't matter for zero vs nonzero, only the sign):

$$A = \begin{bmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{bmatrix}$$

A set of 3 vectors in  $\mathbb{R}^3$  is linearly independent **iff**  $\det(A) \neq 0$ .

Source: standard theorem in linear algebra

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### Step 2: Compute the determinant (cofactor expansion along row 1)

$$\det(A) = 7 \begin{vmatrix} 5 & 0 \\ 7 & -5 \end{vmatrix} - (-8) \begin{vmatrix} -4 & 0 \\ -6 & -5 \end{vmatrix} + 7 \begin{vmatrix} -4 & 5 \\ -6 & 7 \end{vmatrix}$$

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### Step 3: Compute each $2 \times 2$ minor

1.

$$\begin{vmatrix} 5 & 0 \\ 7 & -5 \end{vmatrix} = (5)(-5) - (0)(7) = -25$$

2.

$$\begin{vmatrix} -4 & 0 \\ -6 & -5 \end{vmatrix} = (-4)(-5) - (0)(-6) = 20$$

3.

$$\begin{vmatrix} -4 & 5 \\ -6 & 7 \end{vmatrix} = (-4)(7) - (5)(-6) = -28 + 30 = 2$$

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### Step 4: Put it together (this is where most mistakes happen)

$$\det(A) = 7(-25) - (-8)(20) + 7(2)$$

$$= -175 + 160 + 14 = -1$$

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## Conclusion

$$\det(A) = -1 \neq 0$$

So the vectors are **linearly independent**.

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## Where algebra usually goes wrong

- Forgetting the **minus sign** in cofactor expansion
  - Dropping parentheses:  
 $-(-8)(20)$  mistakenly treated as  $-8(20)$
  - Small  $2 \times 2$  arithmetic error (especially the third minor)
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## Source

- Determinant test for linear independence: Lay, *Linear Algebra and Its Applications*, Section on determinants and linear independence



Why did we choose cofactor expansion instead of minor (i.e. deleting)? That's where I went wrong

**TLDR:** You didn't choose the *wrong* method. You mixed up what "minors" do by themselves versus how they are actually used inside a determinant. Deleting rows/columns alone is not enough — **cofactor signs must be included**. That missing sign is almost certainly where your  $-1$  turned into the wrong answer.

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**Key distinction (this is the whole issue)****Minor  $M_{ij}$ :**Delete row  $i$ , column  $j$ , then take the determinant of what's left.**Cofactor  $C_{ij}$ :**

$$C_{ij} = (-1)^{i+j} M_{ij}$$

**Determinant uses cofactors, not bare minors.**

Source: standard determinant definition

**What you likely did (and why it failed)**

Matrix:

$$A = \begin{bmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{bmatrix}$$

If you "just delete" and multiply:

$$7(-25) + (-8)(20) + 7(2)$$

you get:

$$-175 - 160 + 14 = -321$$

That's **not** a determinant — it's missing the cofactor signs.**What the determinant actually requires**

Cofactor sign pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Source: cofactor sign rule

Row 1 expansion:

$$\det(A) = +7M_{11} - (-8)M_{12} + 7M_{13}$$

So the **middle term flips sign**, which is the entire difference:

$$7(-25) - (-8)(20) + 7(2) = -1$$

## Why cofactor expansion was chosen

Because:

- Determinants are **defined** via cofactor expansion
- "Deleting" (minors) is only an intermediate step
- Cofactors encode orientation via signs

You can expand along **any row or column**, but you *must* use cofactors.

Source: determinant expansion theorem

## Bottom line (robot version)

- Minors  $\neq$  determinant
- Cofactors = minors  $\times$  sign
- You skipped the sign
- That flips the result from wrong to  $-1$

## Sources

- Lay, *Linear Algebra and Its Applications*, Determinants chapter
- Strang, *Introduction to Linear Algebra*, Cofactor expansion definition



it would be helpful if you could tldr explain the process of cofactor expansion + minor and when its appropriate to apply minor during cofactor expansion, which to lead with, etc.

**TLDR:**

You never "choose between" minors and cofactors. **Every determinant expansion is cofactor expansion.** A **minor** is just a sub-step used *inside* cofactor expansion. If you lead with minors alone, the determinant will be wrong because signs are missing.

**Determinant logic (robot version)****1. What the determinant actually is**

For any row  $i$  (or column  $j$ ):

$$\det(A) = \sum_j a_{ij} C_{ij} \quad \text{or} \quad \det(A) = \sum_i a_{ij} C_{ij}$$

This is the **definition** of the determinant.

Source: Lay, *Linear Algebra and Its Applications*, Determinants

**2. What a cofactor is**

$$C_{ij} = (-1)^{i+j} M_{ij}$$

So:

- **Minor**  $M_{ij}$ : delete row  $i$ , column  $j$ , take determinant
- **Cofactor**  $C_{ij}$ : minor  $\times$  checkerboard sign

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**3. Correct workflow (always this order)**

1. **Pick a row or column** (prefer zeros)
2. For each entry  $a_{ij}$ :
  - Compute the **minor**  $M_{ij}$
  - Multiply by the **sign**  $(-1)^{i+j} \rightarrow$  cofactor
  - Multiply by the entry  $a_{ij}$

### 3. Add the results

You **never** stop at minors.

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### 4. Why minors alone are insufficient

Minors measure area/volume **without orientation**.

Determinants need orientation, which is encoded by the sign  $(-1)^{i+j}$ .

If you use minors directly:

$$\sum a_{ij} M_{ij}$$

you are **not computing a determinant**.

Source: determinant orientation property

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### 5. When minors appear

Minors appear **only**:

- As an intermediate calculation
- Inside the formula for cofactors
- Never as the final weighting

So the rule is:

**Lead with cofactor expansion.**

**Use minors only to compute cofactors.**

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### 6. Common failure mode (yours)

You did:

- delete → determinant → multiply → add

But skipped:

- $(-1)^{i+j}$

That flips signs and breaks the determinant.

## One-line memory rule

Determinants use cofactors. Cofactors use minors. Minors alone do nothing.

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### Sources

- Lay, *Linear Algebra and Its Applications*, Sections on determinants and linear independence
- Strang, *Introduction to Linear Algebra*, Cofactor expansion



so for this specific problem, I will be choosing expansion on the columns. So the formula

$\sum_i$  is the one I choose, correct?

**TLDR:** Yes. If you expand down a **column**, you use the  $\sum_i a_{ij}C_{ij}$  formula. Same determinant, different indexing.

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### Determinant formulas (both are valid)

- Row expansion (row  $i$ )

$$\det(A) = \sum_j a_{ij}C_{ij}$$

- Column expansion (column  $j$ )

$$\det(A) = \sum_i a_{ij}C_{ij}$$

These are equivalent definitions of the determinant.

Source: Lay, *Linear Algebra and Its Applications*, Cofactor Expansion Theorem

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### Applied to *this* problem

Matrix:

$$A = \begin{bmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{bmatrix}$$

If you choose **column 2** ( $j = 2$ ):

$$\det(A) = a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32}$$

So yes  $-\sum_i$  is the correct summation because you are moving *down a column*.

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### What does *not* change

- The determinant value
- The cofactor formula  $C_{ij} = (-1)^{i+j}M_{ij}$
- The requirement to include the sign

Source: Strang, *Introduction to Linear Algebra*, Cofactor Expansion

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### Robot rule (lock this in)

- Fix a **column**  $\rightarrow$  sum over **rows** ( $\sum_i$ )
  - Fix a **row**  $\rightarrow$  sum over **columns** ( $\sum_j$ )
  - Always use **cofactors**, never bare minors
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### Sources

- Lay, *Linear Algebra and Its Applications*, Cofactor Expansion Theorem
- Strang, *Introduction to Linear Algebra*, Determinants by Cofactors



